



Development and implementation of a parallel ankle rehabilitation robot:
PRANK

Ante-project presented by:
Luis Fernando Salamanca Sánchez

Supervised by:
Ruben Dario Hernandez Beleno
Mauricio Felipe Mauledoux Monroy

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PROPOSAL SUMMARY SHEET

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Table 1: Proposal Summary Table

3. Author data:

Student Code	Name	ID
3900317	Luis Fernando Salamanca Sánchez	1002460999
Phone	E-mail	
+573508501607	est.luisf.salamanca@unimilitar.edu.co	

Table 2: Author data

Abstract

Human gait is a fundamental component of mobility, autonomy, and overall quality of life. Among the joints involved in locomotion, the ankle plays a crucial role in absorbing shock during heel strike and facilitating propulsion during the push-off phase. Impairments in ankle function—whether due to neurological conditions, trauma, or post-surgical recovery—can significantly compromise gait, balance, and independence. Robotic rehabilitation has emerged as a promising alternative to traditional therapy, offering consistent, repeatable, and quantitatively measurable support.

This project proposes the development of a Parallel Ankle Rehabilitation Robot (PARR), actuated through an RRR electric motor configuration and controlled via an admittance algorithm based on real-time force feedback. The system is complemented by a serious game implemented in virtual reality, designed to enhance patient engagement and tailor therapy difficulty dynamically based on ankle range of motion (ROM) and user-applied force. The project focuses on the mechatronic development of the robot and the comparative validation of its ROM measurement accuracy against an optoelectronic motion capture system, aiming to demonstrate the system's feasibility for reliable biomechanical evaluation and future deployment in clinical environments.

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Chapter 1

PROJECT JUSTIFICATION

Gait is a fundamental aspect of human independence and well-being. Within the gait kinetics, the ankle joint plays a pivotal role in absorbing impact forces during heel strike and generating propulsion during toe-off. When the ankle’s function is impaired—due to conditions such as stroke, orthopedic trauma, or neurodegenerative disease—patients face significant limitations in mobility, balance, and daily autonomy.

Stroke remains one of the leading causes of death and disability worldwide, with approximately 15 million new cases reported each year [1]. The World Stroke Organization emphasizes that this burden continues to rise, particularly in low- and middle-income countries, where access to prevention and treatment is limited. In Colombia, stroke accounts for approximately 120,000 cases and 30,000 deaths annually [2], representing a significant public health challenge that demands effective and accessible rehabilitation strategies.

One of the most debilitating consequences of stroke and other neurological conditions is foot drop—a symptom rather than a disease in itself, arising from multiple underlying pathologies. The most common cause is injury or compression of the peroneal nerve, often linked to trauma, fractures, or prolonged external pressure. Lumbar radiculopathy, particularly involving the L5 nerve root, is another frequent source, typically due to herniated discs or spinal degeneration. Neurological disorders such as stroke, multiple sclerosis, cerebral palsy, and amyotrophic lateral sclerosis can impair motor pathways and lead to foot drop. Muscle-related conditions, including muscular dystrophy and inflammatory myopathies, may also weaken the dorsiflexor muscles. Additionally, systemic diseases like diabetes mellitus contribute through peripheral neuropathy, while tumors, cysts, or autoimmune disorders can cause nerve compression or inflammation. Overall, foot drop reflects a diverse set of neurological, muscular, and systemic pathologies that require careful diagnostic evaluation [3], [4], [5], [6], [7]. Table 1.1 summarizes the epidemiological burden of its primary causes at both the global and national levels, illustrating the scale of the problem and the gaps in local data that motivate targeted intervention.

Foot drop affects approximately 20% of hemiplegic stroke survivors [11], [12], making it one of the most prevalent gait impairments in the post-stroke population. Extrapolating from Colombia’s estimated 120,000 annual stroke cases [2], this proportion implies up to **24,000 new foot drop cases per year** from stroke alone — a figure that does not account for cases arising from diabetes, nerve injury, or neurodegenerative disorders. At the global level, the cumulative burden across all aetiologies is substantially larger, yet epidemiological data specific to foot drop remain fragmented and largely absent from low- and middle-income country

Cause	Global Data	Colombia Data
Stroke (ACV)	15M cases annually; 5M disabled; 5M deaths	120,000 cases/year; 30,000 deaths
Diabetes-related neuropathy	537M adults with diabetes; 30–50% neuropathy	8–9% adult prevalence; neuropathy common
Peroneal nerve injury	Up to 15% of tibia/knee fractures	Reported in trauma hospitals; no national stats
Neurological disorders (MS, ALS, CP)	MS: 2.8M cases; ALS: 1–2/100,000 incidence	Limited data; hundreds of cases in centers
Muscular disorders (dystrophy, myopathies)	Rare; documented globally in thousands	Very limited data; rare case reports

Table 1.1: Global and Colombian Statistics Related to Foot Drop [1], [2], [8], [9], [10]

registries, underscoring the need for accessible, evidence-based rehabilitation technologies in these settings.

Given the scale and diversity of conditions that cause foot drop, effective rehabilitation of the ankle joint is a critical clinical need. However, conventional methods often lack the consistency, objectivity, and sustained engagement required for optimal recovery. Furthermore, existing robotic devices for gait rehabilitation typically do not include mechanisms specifically designed for targeted ankle strength recovery. In this context, robotic rehabilitation systems offer the advantages of repeatability, precise measurement, and the ability to adapt therapy to an individual’s progress. Additionally, mixed reality and serious game interfaces have shown considerable promise in enhancing motivation and participation — key factors that directly influence rehabilitation outcomes [13].

This project addresses these needs through the development of PRANK (Parallel Robot for ANKle rehabilitation), a parallel robotic platform actuated by an RRR configuration and controlled through admittance strategies. It aims to offer a safe, adaptive, and engaging therapeutic environment. By integrating a mixed reality serious game and a real-time feedback loop that adjusts difficulty based on force and range of motion (ROM), the system seeks to promote user-centred recovery while providing measurable, high-fidelity data. To further enhance motor learning and patient engagement, PRANK incorporates **multimodal biofeedback**: visual screen alerts signal performance thresholds and game events, spatial audio cues reinforce correct movement patterns, and vibrotactile feedback provides immediate haptic confirmation of target achievement. This combination of sensory channels has been shown to accelerate motor relearning in neurological rehabilitation [13]. Additionally, validating PRANK’s ROM measurement capabilities against an optoelectronic motion capture system will support its use as a reliable biomechanical assessment tool in future clinical settings.

Chapter 2

STATE OF THE ART

The ankle joint complex is one of the most mechanically demanding structures in the human body. Composed of the tibiotalar, talocalcaneal, and transverse-tarsal joints, it enables three rotational degrees of freedom — dorsiflexion/plantarflexion, inversion/eversion, and adduction/abduction — while simultaneously bearing the full body weight during locomotion [14]. As Morris described as early as 1977, the ankle serves simultaneously as a shock absorber at heel contact, a rigid propulsive lever at push-off, and a mechanism for absorbing limb rotation throughout the stance phase [15]. Disruption of any of these functions causes compensatory gait deviations that increase metabolic cost, reduce stability, and elevate fall risk. Lin et al. quantified these effects in 68 post-stroke patients, demonstrating that dorsiflexor muscle strength is the single strongest predictor of gait velocity ($R^2 = 0.30$) and temporal symmetry ($R^2 = 0.36$), while dynamic plantarflexor spasticity drives spatial asymmetry [16]. Normal ranges of motion span approximately 20–30° dorsiflexion, 40–50° plantarflexion, up to 40° inversion, and 15° eversion — values that directly constrain the workspace requirements of any ankle rehabilitation device.

Among the conditions that compromise ankle function, foot drop is one of the most debilitating. It is the inability to perform adequate dorsiflexion during the swing phase of gait, arising from upper motor neurone lesions — most commonly stroke, multiple sclerosis, or spinal cord injury above T12 — and affecting approximately 20% of stroke survivors with lasting consequences [17], [18]. The resulting compensatory strategies — circumduction, hip-hitching, and steppage gait — are energetically costly and fall-promoting. Conventional management relies on ankle-foot orthoses (AFOs) and functional electrical stimulation (FES), both of which are primarily orthotic rather than rehabilitative. A meta-analysis of eight randomised controlled trials comparing FES and AFO found equivalent improvements in walking speed for both devices, but identified a critical lack of evidence on kinematic and kinetic outcomes and on whether device-induced gains transfer to everyday settings [19], [20]. This gap motivates the development of active robotic systems capable of delivering repeatable, quantitatively measurable, and adaptive therapy. A long-term study of the NESS L300 FES neuroprosthesis in 16 individuals with chronic hemiparesis demonstrated that consistent use over one year improved 10 m walk velocity from 0.67 m/s to 1.06 m/s, with a 23.8% carry-over effect without the device — evidence that sustained ankle assistance can induce neuroplastic changes [17].

Robotic systems have emerged as the most promising platform for delivering this type of therapy consistently, precisely, and adaptively. Among the architectures explored, parallel

robots stand out for their rigidity, high load capacity, and ability to produce multi-degree-of-freedom torques simultaneously — properties that align naturally with the complex kinematics of the ankle [21], [22], [23], [24], [25], [26]. The earliest and most influential series of parallel ankle rehabilitation robots was developed by Jamwal and colleagues at the University of Auckland, beginning with a soft parallel robot (SPR) driven by four pneumatic air muscles [21]. Workspace analysis based on Jacobian singularity criteria and a global conditioning number was performed, and a modified genetic algorithm was used to achieve near-unity kinematic isotropy across the feasible workspace. Subsequent work addressed the forward kinematics problem using a modified fuzzy inference system that outperformed Newton–Raphson iteration in computation speed, making real-time task-space control feasible [21]. A three-stage multi-criteria optimisation framework later decomposed the design problem into kinematic, actuation, and structural stages, applying NSGA-II to simultaneously balance six competing objectives including workspace isotropy, cable tension, and structural compliance [22]. This methodology was further refined with a fuzzy-dominated sorting evolutionary algorithm that improved discrimination among Pareto solutions, providing a reusable template for the dimensional synthesis of future ankle robots [23].

In parallel, Tsoi and Xie designed a three-DoF redundantly actuated platform and demonstrated through singularity analysis that redundant actuation is necessary to avoid singular configurations within the clinical ankle range of motion [14]. Wang et al. extended this direction by performing kinematic simulation of a parallel ankle robot and showing that three-input actuation can cover dorsiflexion $0\text{--}30^\circ$, plantarflexion $0\text{--}50^\circ$, inversion/eversion up to 40° , and adduction/abduction up to 30° [24]. Optimisation of actuator sizing was addressed using a modified differential evolution algorithm with a novel redundant-actuation torque index, outperforming competing methods in global search capability [24]. Spherical parallel mechanisms based on 3-RRS chains have also been explored as an alternative topology: Du et al. obtained eight inverse kinematics solutions analytically and demonstrated accommodation of most patients through an adjustable expansion device [25], while Zuo et al. validated a 3-RRS prototype achieving maximum trajectory errors of 2.5 mm and angular errors within 2° [26].

A comprehensive review by Miao et al. of platform-based ankle rehabilitation robots from 1980 to 2016 concluded that alignment of the robot’s rotation centre with the anatomical ankle joint centre, appropriate workspace, and reconfigurability are the three most critical design requirements — and that parallel mechanisms best satisfy all three [27]. Li et al. subsequently systematised this finding across 65 articles covering 16 unique parallel ankle robots, identifying rotation-centre alignment, three-DoF coverage, and compliance control with variable parameters as the most pressing open challenges, and noting that clinical trials remain scarce [28]. More recently, a 2012–2023 systematic review by Abarca and Elias covering 93 papers confirmed that technology readiness levels remain predominantly at prototype stage and that clinical evidence is limited, underscoring the need for rigorous human-subjects testing [29]. A 2025 review of state-of-the-art ankle rehabilitation robotic devices compared parallel, serial, and exoskeleton topologies across design, actuation, and control dimensions, positioning parallel platforms as the most capable architecture for multi-DoF targeted therapy.

The control of ankle rehabilitation robots has evolved from simple position tracking to biologically inspired compliance strategies that actively recruit patient effort. Impedance control was among the first paradigms applied to parallel ankle platforms: Tsoi and Xie used it to regulate both force and motion simultaneously on their redundantly actuated device,

exploiting the redundant DoF to control the vertical ankle reaction force — a safety-critical quantity [14]. Jamwal et al. later formalised a four-mode impedance control framework (position control, zero-impedance, and two non-zero impedance levels) and demonstrated through experiments with ten neurologically intact subjects that increasing robotic compliance promotes greater active patient participation, a key neuroplasticity-promoting principle [23]. This was extended to an adaptive version that modifies target impedance in real time based on the patient’s active torque contribution: as participation increases the robot reduces its assistance, and as it decreases the robot provides more support [23]. A parallel and arguably more mature development is the compliant ankle rehabilitation robot (CARR) by Zhang et al., which combines a joint-space position controller with a high-level admittance controller in task space and implements a variable-parameter admittance mode that adapts the prescribed trajectory to real-time ankle measurements, achieving maximum NRMSD below 5.4% [25]. A subsequent cascade controller with joint-space force distribution ensured all four pneumatic muscles remained in tension throughout the workspace and achieved NRMSD below 2.3% on an injured human ankle [26]. Li et al. implemented a simpler torque-threshold intent detection strategy on a 2-UPS/RRR platform and confirmed small trajectory errors in passive mode and smooth, responsive motion in active mode, providing a lower-complexity baseline for comparison [30]. Most recently, a 2024 cable-driven parallel ankle robot implemented combined position and admittance control for full-cycle rehabilitation, enabling large-angle multi-DoF coupled training with patient-interaction forces modulating the terminal position in real time

Advanced control strategies such as adaptive impedance control, fuzzy logic, and patient-cooperative control have been developed to enhance engagement and safety across the field [23], [25]. EMG-based intent recognition is increasingly integrated to further personalise therapy: Zaffir et al. compared ANFIS, MLP, LSTM, and combined neural network models for estimating dorsiflexion from only two sEMG channels, finding that LSTM achieved highest accuracy and that adding the average rate of change of EMG consistently improved estimation [31]. These results demonstrate that intent detection from tibialis anterior and gastrocnemius sEMG is feasible with a minimal sensor footprint — enabling a low-cost active control mode complementary to admittance-based therapy. Kitatani et al. added a clinically important nuance: elevated ankle muscle co-activation index during single support in hemiplegic patients was negatively correlated with dorsiflexion angle, plantarflexion moment, and ankle power generation, suggesting that EMG-based co-activation monitoring could serve as a real-time biomarker of paretic instability to guide adaptive parameter adjustment [32]. Cable-driven exoskeletons powered by series elastic actuators offer compliant assistance while enhancing gait symmetry [33], and soft robotic platforms such as T-FLEX integrate modular actuation and flexible interfaces for post-stroke rehabilitation [34]. A soft wearable ankle exoskeleton targeting both dropfoot and excessive inversion during swing was published in 2024, illustrating the complementary roles of wearable ambulatory devices and platform-based clinical robots. In the same year, the AMBLE exoskeleton demonstrated through a nine-week clinical study with chronic hemiparetic stroke patients that adaptive impedance control produces durable improvements in toe clearance and dorsiflexion angular velocity — establishing a direct performance benchmark for PRANK’s future clinical evaluation

Clinical evidence for robot-assisted ankle therapy, while still growing, increasingly supports its integration into standard rehabilitation protocols. A case report of hybrid FES-robotic training in a chronic stroke patient with foot drop over 16 sessions demonstrated improvements in the Functional Ambulation Category and Fugl-Meyer Lower Extremity scores, highlighting

the cognitive engagement produced when patients voluntarily trigger movement onset [20]. Evaluation of Lokomat gait training with kinematic constraints in 15 hemiparetic patients showed significant improvements in peak knee flexion persisting for at least 20 minutes post-session, suggesting that even short bouts of constrained robotic training produce rapid motor learning [17]. Studies comparing robotic therapy against AFO and FES interventions confirm that robot-assisted systems can match or exceed conventional outcomes in gait restoration and muscle activation [19], [20]. A 2023 multi-DoF reconfigurable parallel ankle robot with mechanically adjustable workspace reported clinical improvements in gait symmetry and muscle activation in post-stroke patients, providing a directly comparable topology and target population benchmark for PRANK. The ROM measurement validation component of PRANK’s design — comparing robot-estimated ankle kinematics against an optoelectronic reference — follows a methodology established across multiple recent studies. Markerless pose estimation systems validated against optoelectronic systems have reported ICC values above 0.91 for limb joints under full ROM tasks. The OpenCap system, validated for lower-extremity kinematics during dynamic tasks, achieved RMSE of 2.0–10.2° relative to marker-based capture, providing context for PRANK’s H3 target of $< 5^\circ$.

Virtual reality and serious games have progressively emerged as an indispensable complement to robotic hardware in rehabilitation. Systematic reviews highlight their effectiveness in improving postural balance and engagement in patients with neurological disorders [13], and VR-guided active training has been shown to produce superior outcomes to passive trajectory-following through combined tactile and visual feedback that improves ankle strength, flexibility, and balance [30]. Research on hand dominance in VR reach-to-grasp tasks found a significant positive relationship between movement path length and intentional binding — the patient’s implicit sense of agency — for the dominant limb, suggesting that VR rehabilitation interfaces should personalise feedback and difficulty to the affected side to sustain motivation [18]. A 2024 scoping review of serious games for ankle rehabilitation specifically mapped game genres, accessibility requirements, and usability principles, identifying adaptive difficulty scaling and multimodal feedback as the design variables most strongly linked to therapeutic adherence. Augmented reality and serious gaming integrated with a parallel lower-limb robot demonstrated improved ROM outcomes and engagement compared to standard physiotherapy, directly supporting the inclusion of a VR serious game as PRANK’s primary active training mode. Integration with remote monitoring platforms is also being explored to improve continuity of care and enable home-based therapy [13], [29].

Despite this progress, several challenges remain open. Ensuring compact, low-cost designs suitable for domestic use; achieving real-time adaptation to individual patient effort and variability; standardising clinical protocols for robot-assisted therapy; and integrating biosignal feedback with cloud-based analytics are all active research frontiers. Portability and modularity have motivated reconfigurable robots that adapt to different patient profiles and therapy stages [29], while the broader trend toward decentralised, patient-centred rehabilitation continues to drive development of systems that combine robotic hardware, intelligent control, and engaging digital interfaces. Future work should focus on scalable architectures, hybrid actuation combining rigid and compliant elements, and intelligent control systems that fuse biomechanical models, machine learning, and human-centred design to close the gap between laboratory prototypes and clinically validated tools.

Chapter 3

OBJECTIVES

3.1 General

Develop and validate a three-degree-of-freedom parallel ankle rehabilitation robot (PRANK) with an RRR electric motor configuration, implementing admittance-based control and a mixed reality serious game to deliver adaptive, patient-responsive therapy, and demonstrating the system's ROM measurement accuracy through comparison with an optoelectronic motion capture reference.

3.2 Specific Objectives

- Design, fabricate, and assemble a three-degree-of-freedom parallel robotic structure actuated by an RRR electric motor configuration, integrating force and motion sensing to enable safe and controlled ankle rehabilitation movements.
- Develop an admittance-based control system with adaptive resistance modes—passive, assistive, and resistive—that adjusts therapy difficulty in real time based on ankle ROM and user-applied force, complemented by a mixed reality serious game that provides interactive feedback and promotes patient engagement.
- Validate the system's ROM measurement accuracy through a pilot study with healthy subjects, comparing robot-estimated ankle kinematics against a reference optoelectronic motion capture system to demonstrate feasibility for clinical biomechanical assessment.

Chapter 4

DELIMITATION AND SCOPE

4.1 Type of Research

This project is classified as **experimental** research, since it involves the design, construction, and controlled testing of a physical prototype — the PRANK parallel ankle rehabilitation robot — under conditions defined by the investigator. It also has a **descriptive** component, as one of its core objectives is to characterise the kinematic measurement performance of the system by comparing robot-estimated ankle range of motion (ROM) against a validated optoelectronic motion capture reference under controlled conditions. The experimental phase will determine whether the system meets the performance thresholds stated in the project hypotheses (H1, H2, H3), while the descriptive phase will document the system’s behaviour across its three rehabilitation modes.

4.2 Delimitation Conditions

Space

All design, fabrication, assembly, software development, and experimental testing activities will be carried out at the facilities of the **Universidad Militar Nueva Granada (UMNG)**, Bogotá, Colombia. Specifically, the mechatronics and robotics laboratories of the Faculty of Engineering will be used for mechanical assembly, electronic integration, and hardware-in-the-loop testing. The optoelectronic validation experiments will be conducted in a dedicated biomechanics or motion capture laboratory available at the same institution.

Time

The project will be executed during **2026**, with a planned duration of seven (7) months. The schedule follows the Gantt chart presented in Chapter 8, covering the phases of mechanical design and fabrication, electronic integration, control system implementation, MR interface development, and pilot validation.

Universe

The experimental validation (H3) will be conducted with a sample of **healthy adult subjects**, who will participate in a pilot study to evaluate the robot’s ROM measurement ac-

curacy and the usability of the admittance control system. The use of healthy subjects is appropriate at this stage because the project’s primary goal is technical validation of the system, not clinical assessment of rehabilitation outcomes. Participants with any prior ankle pathology, lower-limb neurological condition, or contraindication to seated robotic exercise will be excluded. The project does not involve clinical trials with patients, and therefore does not require clinical ethics approval beyond institutional research ethics review for non-clinical human-subjects testing.

Scope

The project encompasses:

- **Mechanical:** Design and fabrication of a three-degree-of-freedom parallel robotic structure with an RRR electric motor configuration, covering the therapeutic ankle ROM workspace stated in H1.
- **Electronic and sensing:** Integration of brushless DC motors with encoders, a six-axis load cell for force/torque measurement, and the embedded real-time control platform.
- **Control:** Implementation and experimental validation of an admittance-based control system supporting passive, assistive, and resistive rehabilitation modes (H2).
- **Software:** Development of a mixed reality (MR) serious game in Unity that receives real-time data from the robot and adjusts difficulty based on ankle ROM and applied force.
- **Validation:** A pilot study comparing robot-estimated ankle ROM against an optoelectronic motion capture reference, evaluating the metrics defined in H3.

The project does **not** include:

- Clinical trials with neurological or orthopaedic patients.
- Assessment of long-term rehabilitation outcomes or therapeutic efficacy.
- Evaluation of the psychological or motivational effects of the MR game on patient adherence.
- Commercialisation or regulatory approval activities.

4.3 Research Variables

4.3.1 Independent Variables

Independent variables are those controlled or manipulated by the investigator during the experimental protocol:

- **Control mode:** The rehabilitation mode selected for each trial — passive, assistive, or resistive. This variable determines the role of the robot (guiding, following, or resisting patient motion) and the configuration of the admittance parameters.

- **Admittance parameters (K_d , B_d , M_d):** The virtual stiffness, damping, and inertia matrices of the admittance controller (Equation 6.3). These are set by the operator at the beginning of each session and may be varied systematically across trials to characterise controller behaviour.
- **Prescribed trajectory / ROM target:** The reference ankle trajectory defined for each rehabilitation exercise (e.g., sinusoidal dorsiflexion/plantarflexion at a specified amplitude and frequency). Different trajectory profiles constitute different experimental conditions.
- **VR game difficulty level:** The amplitude and speed demands of the virtual task presented to the subject, which modulates the motor challenge independently of the robot’s physical resistance.

4.3.2 Dependent Variables

Dependent variables are those measured as outputs of the system in response to the independent variable conditions:

- **Ankle range of motion (ROM):** The angular excursion of the ankle in dorsiflexion/-plantarflexion, inversion/eversion, and internal/external rotation, estimated from the robot’s encoders. This is the primary variable for H3, compared against the optoelectronic reference.
- **Interaction force and torque:** The six-component wrench measured by the load cell at the robot-foot interface, in Newtons and Newton-metres. This variable is the input to the admittance controller and the primary observable for evaluating H2.
- **Controller tracking error (RMSE):** The root mean square deviation between the prescribed reference trajectory and the measured ankle trajectory, expressed in degrees. Used to quantify control performance in passive and assistive modes.
- **Controller settling time:** The time required for the system to reach steady-state response after a step change in reference position or admittance parameters, in milliseconds. Used to quantify controller responsiveness (H2).
- **ROM measurement error vs. optoelectronic reference:** The difference between robot-estimated and optoelectronic-measured ROM, expressed as RMSE (degrees) and Pearson correlation coefficient. The primary metric for H3.

4.4 Limitations

The following limitations are acknowledged and will be addressed in the interpretation of results:

- **Healthy subjects only:** The pilot study uses neurologically intact adults. Differences in ankle stiffness, spasticity, and motor variability between healthy and patient populations may affect the generalisability of ROM accuracy and controller performance results to clinical settings.

- **Laboratory conditions:** All tests are conducted under controlled laboratory conditions (fixed seating, supervised sessions, controlled lighting for optoelectronic capture). Performance in real-world or home-based settings may differ.
- **Prototype stage:** The robot is a research prototype; results reflect the performance of this specific implementation and may not generalise to future iterations with different actuators, sensors, or structural materials.
- **Short time horizon:** The seven-month duration precludes longitudinal assessment of motor learning, fatigue effects, or inter-session variability in the admittance controller's adaptive behaviour.
- **VR validation limited to usability:** The VR serious game will be evaluated for usability and engagement, but its therapeutic efficacy as a rehabilitation tool is outside the scope of this project.

Chapter 5

HYPOTHESIS

H1 — Mechanical Design: The PRANK parallel robotic structure will achieve a functional workspace covering at least $\pm 25^\circ$ of dorsiflexion/plantarflexion, $\pm 15^\circ$ of inversion/eversion, and $\pm 10^\circ$ of internal/external rotation — values established as the minimum therapeutic range of motion required for ankle rehabilitation [14], [35].

H2 — Control System: The admittance-based control system will demonstrate stable and responsive behaviour across passive, assistive, and resistive rehabilitation modes, with a settling time under 500 ms and a force tracking error below 15% of the commanded torque, ensuring safe and adaptive interaction with the patient throughout therapy [25], [30].

H3 — ROM Validation: Ankle ROM measurements obtained from the PRANK robot will show a Pearson correlation coefficient above 0.95 and a root mean square error below 5° compared to those captured by a validated optoelectronic motion capture system, demonstrating the feasibility of the system as a reliable biomechanical assessment tool in an ankle rehabilitation context.

Chapter 6

THEORETICAL FRAMEWORK

6.1 Human Gait

Human gait is one of the most important tasks of daily life, as it allows people to maintain independence and live with autonomy, mobility, and dignity. Impairments in gait due to neurological or musculoskeletal conditions can significantly affect quality of life, making rehabilitation a critical component of recovery and functional reintegration.

The gait analysis is carried out on the basis of the gait cycle, which can be taken as a representation of a person's walking patterns; comparison of several cycles indicates the variability of the pattern [17]. The gait cycle is normalized from 0% to 100% because each individual pattern has different absolute timing, making direct cycle-length comparisons non-objective [18]. It can be described with spatial and temporal parameters.

6.1.1 Spatial parameters

The main spatial parameters are shown in Figure 6.1 and described below:

- **Step:** the movement of one foot in front of the other.
- **Stride:** a step of one foot followed by a step of the other foot; one complete gait cycle.
- **Foot contact:** the beginning of the gait cycle, referred to as *heel strike* in healthy individuals.
- **Step length:** distance traveled by one foot in front of the same reference point of the other foot.
- **Stride length:** distance between two consecutive heel strikes of the same foot.
- **Step width:** mediolateral separation of the feet, also known as stride width.

6.1.2 Temporal parameters

Temporal parameters are listed below:

- **Stride time:** duration of one gait cycle (time between two consecutive foot strikes of the same foot).

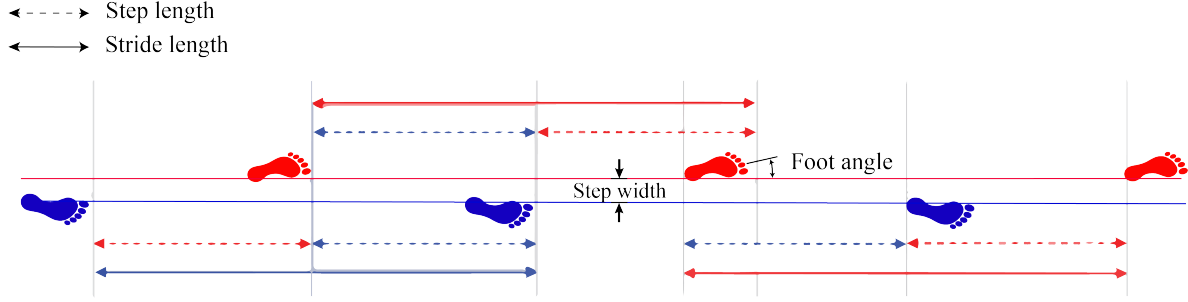


Figure 6.1: Gait spatial parameters. Source: edited by author from [17].

- **Cadence:** a commonly used indirect measure of cycle frequency, expressed as:

$$\text{Cadence} = \frac{\text{number of cycles}}{\text{time interval}} \quad (6.1)$$

- **Walking speed:** distance traveled per unit time in m/s. If cadence is in steps per minute and stride length is in meters:

$$v = \frac{\text{cadence} \times \text{stride length}}{120} \quad (6.2)$$

Gait is globally divided into two phases: *stance* and *swing*. Stance occurs when the foot is in contact with the ground; swing occurs when it is not. The stance phase ends at toe-off, which is simultaneously the onset of swing, as illustrated in Figure 6.2.

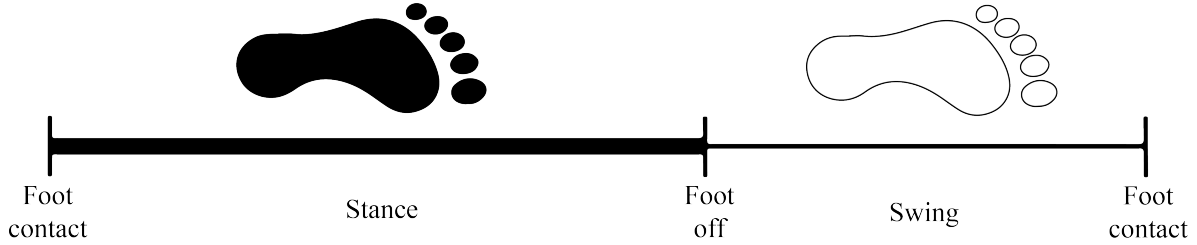


Figure 6.2: One side gait phases. Source: edited by author from [17].

Including both legs, the gait cycle is divided into first double support (both feet in contact), single support, second double support, and swing, as shown in Figure 6.3. Single support and swing are long phases, so each is further subdivided into *early*, *middle*, and *late* sub-phases [17], [36].

Any impairment in the ankle joint manifests as alterations in gait parameters, disrupting the cyclical and repetitive nature of walking. This leads to increased energy expenditure and triggers compensatory mechanisms in other muscle groups, often resulting in instability and a heightened risk of falling.

6.2 Anatomy of the Ankle Joint Complex

The ankle joint complex (AJC) is not a simple hinge joint but a mechanically sophisticated structure composed of three articulations working in concert: the **tibiotalar (talocrural)**

joint, the **talocalcaneal (subtalar) joint**, and the **transverse-tarsal (talocalcaneonavicular) joint** [35]. Together, these joints enable three rotational degrees of freedom that are essential for locomotion.

6.2.1 Tibiotalar (Talocrural) Joint

The talocrural joint is formed by the distal tibia, the distal fibula, and the trochlea of the talus. The medial and lateral malleoli form a mortise that constrains the talus and confers primary stability. Although the joint functions predominantly as a hinge, the conical geometry of the trochlea surface and the oblique orientation of its rotation axis introduce a coupled axial rotation component during dorsiflexion and plantarflexion, making it biomechanically more complex than a pure hinge [35]. The talus is widest anteriorly, so the joint is most congruent and stable in dorsiflexion — a clinically important asymmetry.

Stability is provided by three ligament groups [35], [37]:

- **Lateral collateral ligament complex:** anterior talofibular ligament (ATFL), calcaneofibular ligament (CFL), and posterior talofibular ligament (PTFL). The ATFL is the most frequently injured ligament in ankle sprains.
- **Medial collateral (deltoid) ligament:** a broad, triangular ligament complex that resists eversion and provides medial stability.
- **Tibiofibular syndesmosis:** anterior and posterior tibiofibular ligaments plus the interosseous membrane, maintaining the integrity of the mortise under dynamic loading.

6.2.2 Subtalar (Talocalcaneal) Joint

The subtalar joint is formed by three facets between the talus and calcaneus — posterior, middle, and anterior — and functions as a triplanar, uniaxial joint [35]. Its oblique rotation axis (inclined approximately 42° from horizontal and 16° from the sagittal plane) means that motion at this joint is inherently triplanar: dorsiflexion–abduction–eversion in one direction and plantarflexion–adduction–inversion in the other. This coupling is fundamental to gait: the subtalar joint unlocks during swing to allow shock absorption, and locks during push-off to convert the foot into a rigid lever [35]. The interosseous talocalcaneal ligament is its

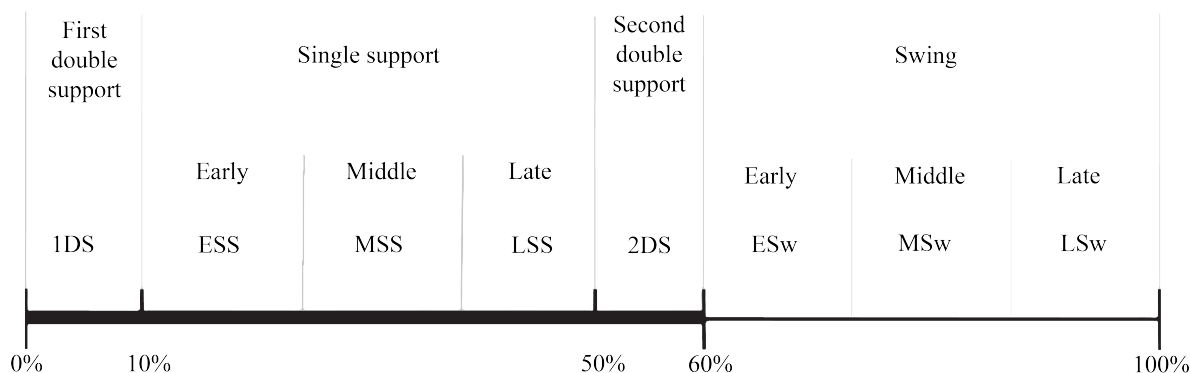


Figure 6.3: Both sides gait phases. Source: edited by author from [17].

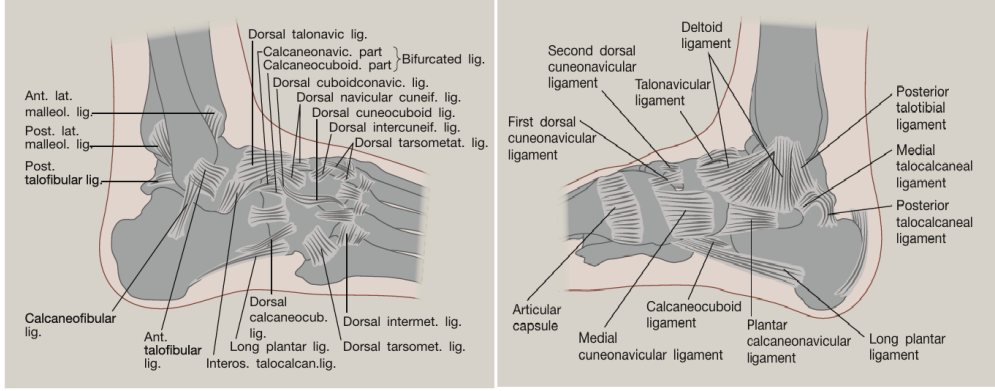


Figure 6.4: Ligament anatomy of the ankle joint complex. Source: author.

primary restraint, complemented by the CFL (the only ligament bridging both the talocrural and subtalar joints) and surrounding tendons.

6.2.3 Muscles and Tendons

The principal muscle groups acting on the ankle are [35], [37]:

- **Dorsiflexors:** tibialis anterior (primary), extensor digitorum longus, extensor hallucis longus. These muscles prevent foot drop during swing and control forefoot landing at heel strike.
- **Plantarflexors:** gastrocnemius and soleus (triceps surae, acting via the Achilles tendon), tibialis posterior, peroneus longus and brevis. The triceps surae generates the propulsive push-off force.
- **Invertors / Evertors:** tibialis posterior (inversion), peroneus longus and brevis (eversion), which contribute to frontal-plane stability and foot posture.

The Achilles tendon, transmitting triceps surae forces to the calcaneus, is the largest and strongest tendon in the body. Its role in energy storage and return during the stance-to-swing transition makes it a key structure for walking efficiency [38], [39].

6.3 Kinematics and Kinetics of the Ankle in Gait

6.3.1 Range of Motion

The clinical and biomechanical literature reports the following reference ranges for the AJC [14], [35], [37]:

During normal walking, the total sagittal-plane excursion is approximately 30° (from $\sim 5^\circ$ dorsiflexion at heel strike to $\sim 20^\circ$ plantarflexion at toe-off). Stair climbing demands up to 37° dorsiflexion and 56° plantarflexion, illustrating the need for wider workspace in rehabilitation devices targeting functional activities beyond level-ground walking. These values directly constrain the workspace requirements of PRANK: the robot must cover at least $\pm 25^\circ$ dorsiflexion/plantarflexion, $\pm 15^\circ$ inversion/eversion, and $\pm 10^\circ$ internal/external rotation to enable comprehensive ankle rehabilitation.

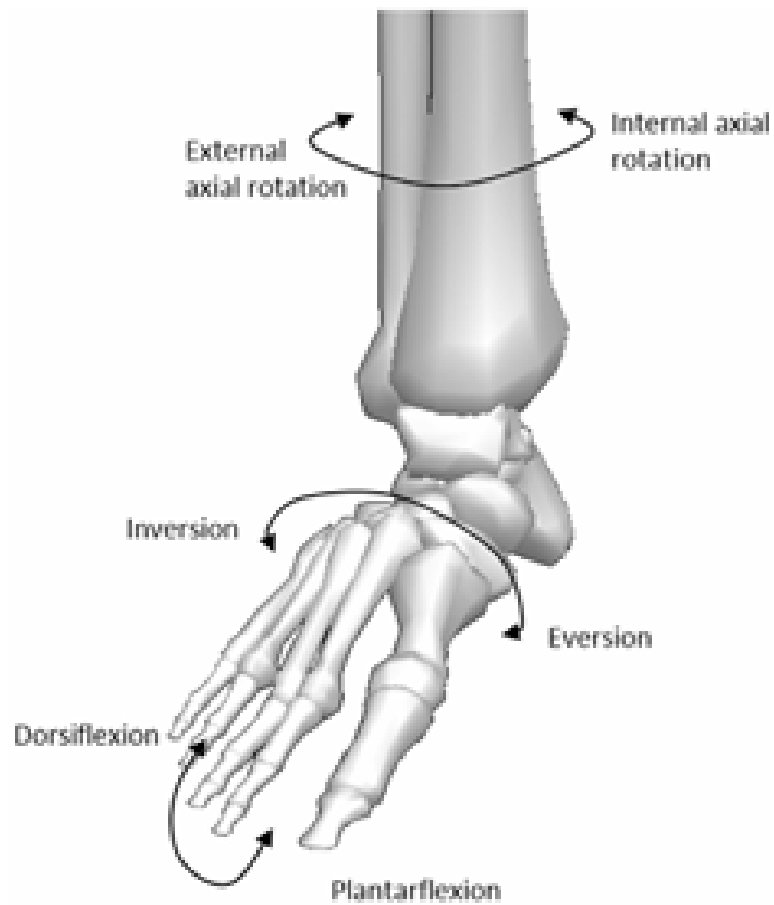


Figure 6.5: Reference planes and angular degrees of freedom of the ankle joint complex.
Source: author.

Table 6.1: Reference ranges of motion of the ankle joint complex

Movement	Full passive ROM	Required for gait
Dorsiflexion	10°–20°	≈ 15°
Plantarflexion	40°–55°	≈ 20°
Inversion	20°–35°	≈ 5°–10°
Eversion	10°–15°	≈ 5°
Internal / External rotation	±5°–15°	Coupled to subtalar motion

6.3.2 Ankle Kinematics During the Gait Cycle

The ankle follows a characteristic kinematic pattern across the gait cycle [35], [37]:

- **Heel strike (0%):** The foot lands in slight plantarflexion ($\sim 5^\circ$); tibialis anterior activates eccentrically to lower the forefoot and control impact.
- **Loading response (0–12%):** Controlled plantarflexion to $\sim 5^\circ$ as the foot flattens to the ground.
- **Mid-stance (12–50%):** Progressive dorsiflexion as the tibia advances over the fixed foot, reaching peak dorsiflexion of $\sim 10^\circ$ – 15° .
- **Terminal stance and pre-swing (50–62%):** Rapid plantarflexion to $\sim 20^\circ$, with the gastrocnemius–soleus generating the propulsive push-off burst.
- **Swing (62–100%):** Dorsiflexion returns to neutral to clear the ground; failure to achieve this is the defining feature of *foot drop*.

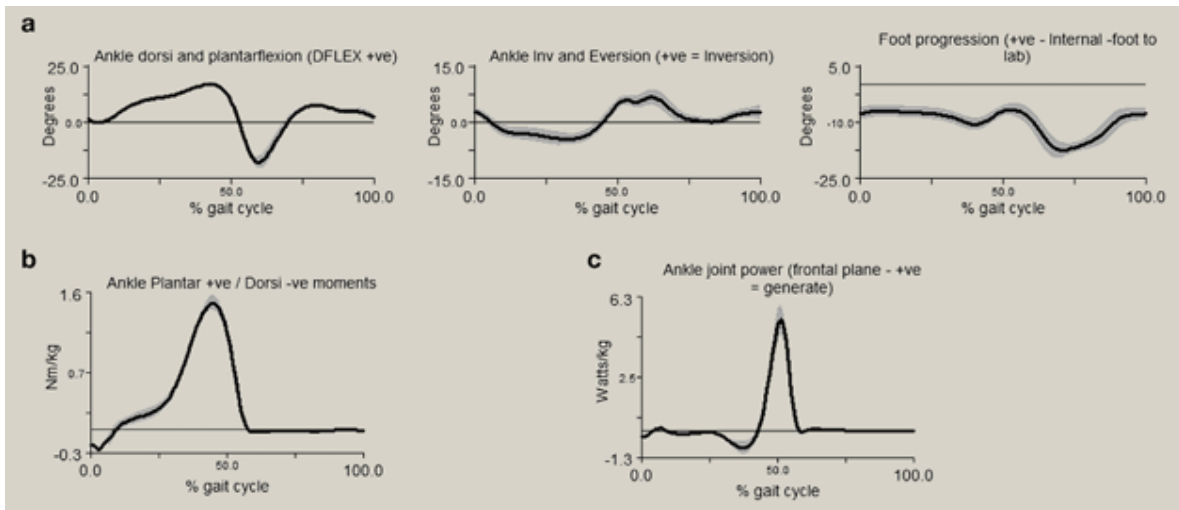


Figure 6.6: Ankle joint kinematics and kinetics during the gait cycle: angular displacement, moment, and power profiles. Source: author.

6.3.3 Ankle Kinetics and Power

The kinetics of the ankle during gait are dominated by two events [37], [39]:

- **H1 — controlled plantarflexion (loading response):** a small energy-absorption event as the tibialis anterior decelerates forefoot drop.
- **A2 — push-off power burst (pre-swing):** the largest energy-generation event in normal gait. Peak plantarflexion torque reaches **1.4–1.6 Nm/kg** of body mass, and peak ankle power reaches **2.5–4.0 W/kg** at approximately 55–60% of the gait cycle [37], [38].

For a 70 kg adult this corresponds to approximately 98–112 Nm peak torque and 175–280 W peak power at push-off — the highest power output of any lower-limb joint during walking. These values are the primary mechanical specifications for PRANK: the robot’s actuators and structural frame must sustain these loads during resistive and assistive training without compromising safety. Table 6.2 summarises the key kinetic parameters:

Table 6.2: Normative ankle kinetic parameters during level-ground walking (70 kg reference subject)

Parameter	Normalised	Absolute (70 kg)
Peak plantarflexion torque (push-off)	1.4–1.6 Nm/kg	98–112 Nm
Peak ankle power (push-off, A2)	2.5–4.0 W/kg	175–280 W
Peak dorsiflexion torque (swing)	<0.2 Nm/kg	<14 Nm
Peak angular velocity (push-off)	~4.3 rad/s	—

6.4 Biomechanical Modelling of the Ankle

Biomechanical models of the ankle range from simplified single-axis representations to high-fidelity multi-segment finite element models [40], [41]. For robot control purposes, the most relevant level of abstraction is the **rigid-body task-space model**: the ankle is represented as a spherical joint with three rotational degrees of freedom, and the interaction between the robot’s footplate and the patient’s foot is described by a wrench (force and torque vector) applied at a defined reference point.

A key insight from the biomechanics literature is that the ankle’s mechanical behavior is not constant: its **quasi-stiffness** — the slope of the torque-angle curve — varies from ~ 2 Nm/° during controlled plantarflexion (loading response) to ~ 8 Nm/° during push-off [38]. This nonlinear, phase-dependent impedance means that fixed-gain position controllers are insufficient for safe human–robot interaction: a controller that is stiff enough to guide passive ROM training would resist voluntary patient effort in active training modes. This is the fundamental motivation for **admittance control**.

6.4.1 Admittance Control and Its Connection to Ankle Biomechanics

Admittance control is a human–robot interaction paradigm in which the robot accepts force or torque inputs from the patient and responds by modifying its motion output [25], [30].

The admittance relationship maps a measured interaction wrench $\mathbf{F}$ (6-dimensional force/-torque vector) to a commanded velocity or position deviation $\dot{\mathbf{x}}$ through a virtual mechanical impedance:

$$M_d \ddot{\mathbf{x}} + B_d \dot{\mathbf{x}} + K_d \mathbf{x} = \mathbf{F} \quad (6.3)$$

where M_d , B_d , and K_d are the desired virtual inertia, damping, and stiffness matrices, respectively. By tuning these parameters, the therapist or the control algorithm can select the robot’s apparent mechanical resistance independently of its physical structure.

The connection to ankle biomechanics is direct:

- **Passive training mode:** K_d is set high and B_d is moderate — the robot guides the ankle along a prescribed trajectory, mimicking the feel of manual physiotherapy.
- **Assistive training mode:** K_d is reduced and M_d is low — the robot follows the patient’s voluntary motion with minimal resistance, encouraging active participation. This exploits the neuroplasticity principle that patient-cooperative training produces greater cortical reorganisation than purely passive movement.
- **Resistive training mode:** B_d is increased — the robot adds a velocity-proportional resistive torque that opposes the patient’s movement, providing progressive strength training analogous to the push-off loading experienced in normal gait.

Adaptive admittance control extends this by varying M_d , B_d , K_d in real time based on patient performance metrics — for example, reducing K_d as the patient’s active torque contribution increases [23], [25]. This directly replicates the role of the ankle’s own variable quasi-stiffness in normal gait, making the robot biomechanically compatible with the natural mechanics of the joint it is rehabilitating.

The biomechanical parameters from Table 6.2 provide concrete bounds for the admittance design: the virtual stiffness K_d must be bounded below by the minimum torque needed to passively move the ankle (approximately 0.2 Nm/° based on cadaveric measurements) and above by the peak torque the device must safely resist (1.6 Nm/kg $\times$ patient mass), while the virtual damping B_d must be sufficient to attenuate oscillations within the ankle’s natural bandwidth of approximately 3–5 Hz during gait.

6.5 Motor Rehabilitation and Neuroplasticity

Motor recovery after injury or neurological impairment relies on neuroplasticity — the brain’s ability to reorganize through repetitive, goal-oriented movement. Rehabilitation robotics leverage this principle by enabling high-intensity, task-specific training with precise, reproducible mechanics. Key motor learning principles include augmented feedback, movement variability, and active participation, all of which enhance cortical reorganisation and functional recovery [42], [43]. PRANK integrates these principles through interactive VR tasks that provide real-time visual and haptic feedback, and through an admittance control architecture that demands and rewards active patient effort.

6.6 Human–Machine Interfaces in Rehabilitation

Effective rehabilitation requires intuitive and responsive interfaces between the user and the robotic system. These interfaces must accommodate physical comfort, cognitive engagement, and safety. Advances in flexible electronics and soft robotics have enabled more ergonomic designs that conform to the user’s anatomy [44]. Virtual reality (VR) environments provide immersive feedback, increasing motivation and facilitating motor learning through gamified rehabilitation tasks [45]. Research shows that combining tactile and visual feedback improves ankle strength, flexibility, and balance beyond what passive trajectory-following achieves, and that personalising the game interface to the affected limb side sustains the patient’s implicit sense of agency [18].

6.7 Clinical Evaluation Metrics

Quantifying rehabilitation outcomes is essential for validating system effectiveness. Clinical metrics include range of motion, gait symmetry, balance, and patient-reported outcomes. Instruments such as the Rehabilitation Measures Database offer standardised tools for assessing ankle functionality and tracking progress over time [46]. In the context of PRANK, primary quantitative outcomes are: (1) ankle ROM across all three axes, validated against an optoelectronic motion capture reference; (2) force tracking error and settling time of the admittance controller; and (3) patient-reported usability and engagement scores from the VR serious game. These metrics are directly linked to the three hypotheses of the project and inform the design of the pilot study protocol.

Chapter 7

MATERIALS AND METHODS

This chapter describes the system architecture and development methodology of PRANK — a Parallel Robot for ANKle rehabilitation. The project is structured in five sequential stages that span mechanical design, electronic integration, control implementation, mixed-reality interface development, and experimental validation. This staged structure supports project scheduling and provides clear deliverables and entry/exit criteria at each phase. Development follows established engineering best practices for mechatronic systems and rehabilitation robotics, with safety and software quality considerations aligned with relevant international standards where applicable.

Figure 7.1 shows the physical schematic of the PRANK robot, and Figure 7.2 presents the overall system architecture, showing the integration between the mechanical platform, embedded hardware, control software, and the mixed-reality interface.

7.1 Stage 1 — Mechanical Design and Fabrication

The PRANK robot is a three-degree-of-freedom (DoF) parallel structure designed to produce controlled rotational motion at the ankle joint in three anatomical planes: dorsiflexion/plantarflexion, inversion/eversion, and internal/external rotation. The kinematic configuration follows an RRR (revolute–revolute–revolute) chain arrangement, in which three independent motor-driven arms converge on a common moving platform — the footplate — through spherical or universal joints.

The mechanical structure is constructed from lightweight aluminium profiles and 3D-printed structural joints, combining rigidity with modularity and ease of component replacement. The footplate includes an adjustable strapping system and a contoured sole support to accommodate different foot sizes and ensure reliable mechanical coupling between the patient’s foot and the robot. Rotation centre alignment with the anatomical ankle joint centre is a primary design constraint, implemented through geometric calibration of the kinematic chain during the design phase.

The full 3D mechanical design is developed in **SolidWorks**, which is also used for structural simulation (FEA) to verify that the frame, joints, and footplate withstand the peak loads reported in the biomechanics literature (up to ~ 112 Nm at the ankle during push-off equivalent loading). Kinematic workspace coverage — at minimum $\pm 25^\circ$ dorsiflexion/plantarflexion, $\pm 15^\circ$ inversion/eversion, and $\pm 10^\circ$ internal/external rotation (H1) — is verified through SolidWorks Motion analysis prior to fabrication.

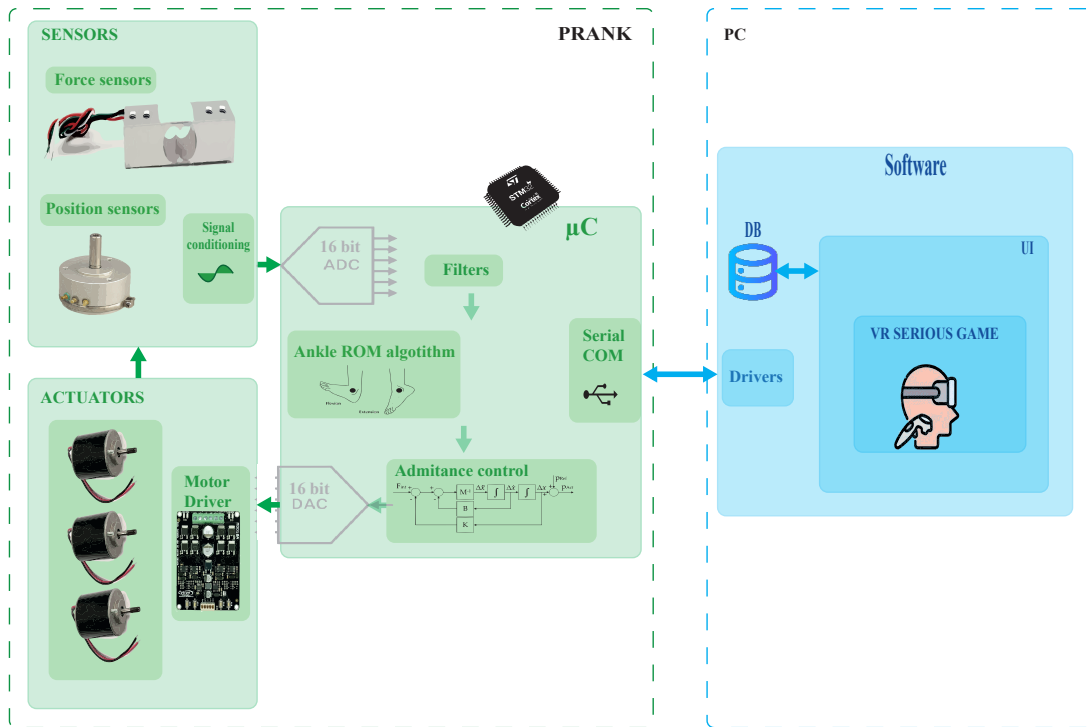


Figure 7.1: PRANK physical system schematic.

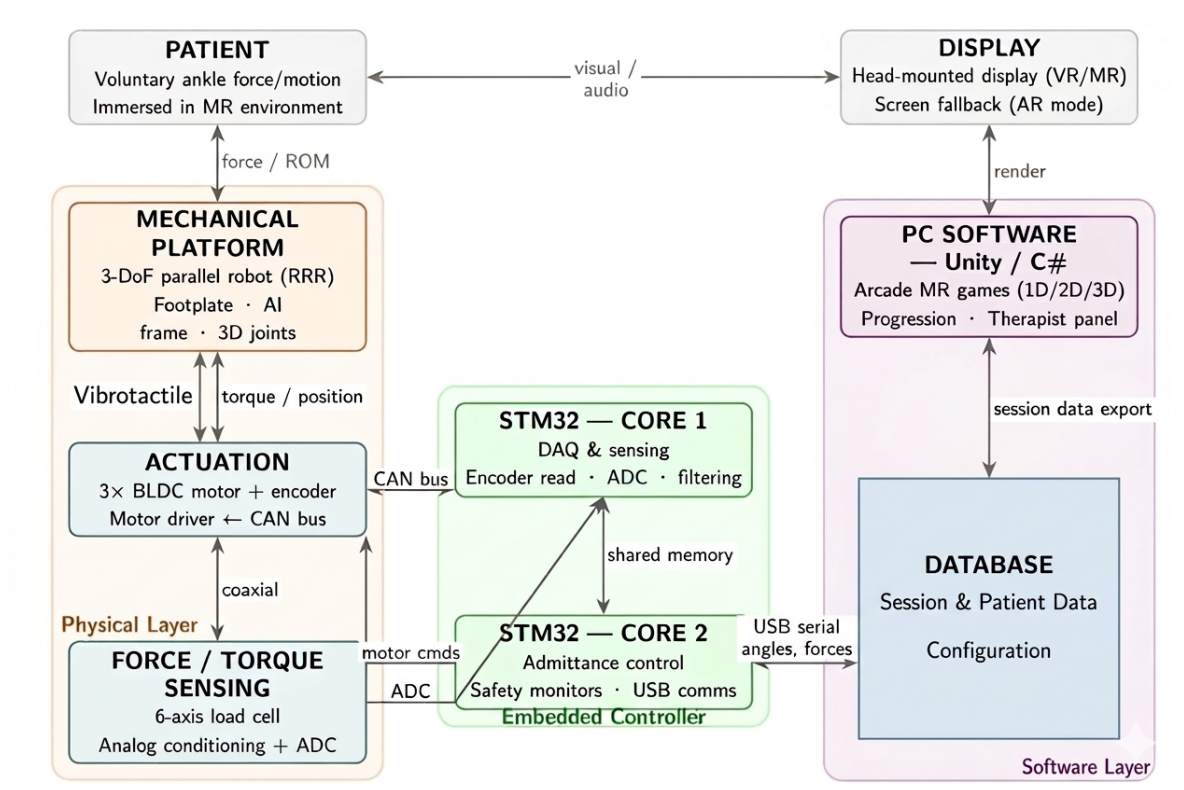


Figure 7.2: PRANK system architecture: physical platform, dual-core embedded controller, and software layer with mixed reality interface.

Stage deliverable: Assembled and mechanically validated parallel robotic platform.

7.2 Stage 2 — Electronic Integration and Sensing

Each of the three kinematic chains is actuated by a **brushless DC motor** with an integrated incremental encoder, providing position feedback with sufficient resolution for ankle-level kinematics ($< 0.1^\circ$ angular resolution at the footplate). Motor drivers communicate with the embedded controller via a CAN bus, providing a noise-robust, real-time communication channel suitable for multi-axis coordinated motion.

Force and torque sensing at the robot–patient interface is performed by a **six-axis load cell** (also referred to as a 6-DoF force/torque sensor) mounted between the footplate and the last structural link. Unlike a 3D force sensor — which measures only the three linear force components (F_x, F_y, F_z) — a six-axis load cell additionally measures the three torque components (M_x, M_y, M_z), providing the full interaction wrench required by the admittance controller to detect rotational intent at the ankle. This distinction is critical: without the torque channels, the controller cannot distinguish whether the patient is attempting dorsiflexion from inversion, making a 6-axis sensor necessary despite its higher cost. Analog signals from the load cell are conditioned and digitised by a dedicated ADC front-end before transmission to the controller.

The embedded processing platform is a dual-core **STM32** microcontroller. The two cores are assigned distinct, real-time tasks to avoid scheduling conflicts and ensure deterministic execution:

- **Core 1 — Sensing and data acquisition:** Reads encoder positions, load cell measurements, and any auxiliary sensors at a fixed sampling rate; performs signal filtering and packaging; and passes processed data to Core 2 via shared memory with a hardware semaphore.
- **Core 2 — Control and communications:** Executes the admittance control law, generates motor torque commands, manages safety monitors (torque limits, emergency stop), and handles serial communication with the Unity game engine on the host PC.

This dual-core partitioning ensures that sensing and control loops run at their required update rates independently, preventing communication overhead from degrading control performance. Firmware is developed in **C/C++** using the STM32 HAL and FreeRTOS for task scheduling.

Stage deliverable: Fully wired and calibrated mechatronic platform with verified sensor readings and motor control.

7.3 Stage 3 — Control System Implementation

The control architecture implements three rehabilitation modes — passive, assistive, and resistive — all built on the admittance control framework described in the theoretical framework (Equation 6.3).

In **passive mode**, the robot executes a predefined reference trajectory (sinusoidal or custom) with high virtual stiffness K_d , guiding the ankle through the target ROM regardless of patient effort. In **assistive mode**, virtual stiffness is reduced and the admittance law allows

the patient’s voluntary interaction forces to modify the prescribed trajectory, encouraging active participation. In **resistive mode**, a velocity-proportional virtual damping B_d opposes the patient’s motion, providing progressive mechanical resistance for strength training.

The admittance control law and mode-switching logic are implemented directly in **C/C++** on the STM32 platform. Controller gains and admittance parameters are tuned experimentally through a systematic parameter sweep on the physical prototype, and performance is assessed against the criteria in H2: settling time < 500 ms and force tracking error $< 15\%$ of commanded torque. Offline data analysis and result plotting are performed in MATLAB.

Safety mechanisms implemented in firmware include: per-axis torque saturation, workspace boundary enforcement, watchdog timer, and a hardware emergency-stop circuit.

Stage deliverable: Implemented, tuned, and safety-validated admittance control system operating in all three rehabilitation modes.

7.4 Stage 4 — Mixed Reality Interface Development

The patient-facing interface is a **mixed reality (MR) serious game** developed in **Unity** with **C#**. The game world is fully virtual — the patient is not shown their foot or the robot — and the ankle motion is mapped to the control of an in-game character or object, creating an intuitive and engaging rehabilitation experience.

The system supports two display modalities:

- **Virtual reality (VR) mode:** full immersive display via a head-mounted display (HMD), providing a completely virtual environment.
- **Mixed reality (MR) mode:** virtual game elements are overlaid on the physical environment using a passthrough HMD, allowing the patient to remain aware of their surroundings while playing.

Game design follows an **arcade paradigm**: fast-paced, visually clear games inspired by classical titles where the ankle controls the game action directly. Three game archetypes are planned, each targeting a different movement dimensionality:

- **1D games** (e.g., Flappy Bird-style): dorsiflexion/plantarflexion only controls altitude or height of an object. Used in early rehabilitation stages or when only one axis has sufficient motor control.
- **2D games** (e.g., car driving, side-scrolling): two ankle axes (e.g., plantarflexion/dorsiflexion + inversion/eversion) control steering and speed simultaneously, requiring coordinated biaxial movement.
- **3D games** (e.g., air hockey, ball rolling): all three ankle axes are mapped to full 3D object control, requiring the most complex and complete ankle motor activation.

Progression between 1D, 2D, and 3D modes is decided by the therapist based on the patient’s ROM and force metrics logged across sessions. The game communicates with the STM32 Core 2 over a serial USB link, receiving real-time ankle joint angles at a rate sufficient for smooth visual feedback (≥ 60 Hz). To reinforce motor learning, the interface implements **multimodal biofeedback** across three sensory channels:

- **Visual alerts:** on-screen indicators (colour changes, flashes, score events) signal when the patient reaches a ROM target, exceeds a force threshold, or completes a game objective.
- **Spatial audio:** directional sound cues tied to game events — success tones, collision effects, rhythm cues — reinforce correct movement patterns and provide timing information independent of visual attention.
- **Vibrotactile feedback:** haptic actuator on the footplate or HMD provides a brief vibration pulse upon target achievement, offering a proprioceptive confirmation channel that remains effective even when the patient’s visual focus is occupied by the game.

A therapist dashboard allows session configuration, mode selection, biofeedback intensity adjustment, and review of performance metrics (ROM achieved, task completion rate, applied force profile).

Stage deliverable: Functional MR serious game with three arcade archetypes (1D/2D/3D), integrated with the robot hardware, and therapist dashboard operational.

7.5 Stage 5 — System Integration and Experimental Validation

The final stage integrates all subsystems and conducts the experimental protocol described in the project hypotheses.

H1 — Workspace validation: The robot is commanded to its mechanical limits in each axis and the achieved angles are measured with the on-board encoders and confirmed visually. The target values ($\pm 25^\circ$, $\pm 15^\circ$, $\pm 10^\circ$) must be reached without binding or motor saturation.

H2 — Controller validation: A series of step-reference and sinusoidal tracking experiments are performed with a mechanical ankle phantom and subsequently with healthy subjects. Force tracking error and settling time are computed from the logged sensor data and compared against the H2 thresholds.

H3 — ROM validation: A pilot study with healthy adult subjects is conducted simultaneously using the robot’s encoder-based ROM estimation and a reference optoelectronic motion capture system. Both systems record ankle joint angles across passive ROM trials. Pearson correlation coefficient and RMSE between the two measurement systems are computed and compared against the H3 thresholds ($r > 0.95$, $\text{RMSE} < 5^\circ$).

All experimental data are logged to file by Core 2 and the Unity dashboard for offline analysis in MATLAB. Results are reported as mean $\pm$ standard deviation across subjects and trials.

Stage deliverable: Experimental dataset and statistical analysis report validating H1 and H3 (H2 is validated at the end of Stage 3).

Chapter 8

SCHEDULE

The development of the PRANK ankle rehabilitation robot is structured into the five stages described in the Materials and Methods chapter. The project timeline spans **7 months** (2026), with parallel execution of independent workstreams where feasible to maximise efficiency within the available time, as illustrated in Figure 8.1.

The schedule is organised as follows:

- **Stage 1 — Mechanical Design and Fabrication (Months 1–3):** 3D design in SolidWorks, structural FEA, kinematic workspace verification via SolidWorks Motion, component procurement, and physical assembly of the parallel platform.
- **Stage 2 — Electronic Integration and Sensing (Months 2–4):** Selection and procurement of BLDC motors, encoders, motor drivers, six-axis load cell, and STM32 board. PCB layout for signal conditioning, wiring, dual-core firmware architecture setup (FreeRTOS, CAN bus, ADC), and sensor calibration. This stage begins in parallel with mechanical design once component specifications are fixed.
- **Stage 3 — Control System Implementation (Months 3–5):** C/C++ implementation of the admittance control law on the STM32, passive/assistive/resistive mode logic, safety monitors, and experimental tuning of admittance parameters. H2 validation experiments (step-reference and sinusoidal tracking) are conducted at the end of this stage.
- **Stage 4 — Mixed Reality Interface Development (Months 2–6):** Unity/C# development of the three arcade game archetypes (1D, 2D, 3D), USB serial integration with the STM32, progression logic, and therapist dashboard. This stage runs in parallel with hardware development and converges at system integration.
- **Stage 5 — System Integration and Experimental Validation (Months 6–7):** Full system assembly and end-to-end testing, H1 workspace validation, H3 ROM pilot study with optoelectronic reference, statistical analysis, and documentation of results.

Continuous **documentation** of design decisions, experimental results, and the final pre-project report runs across all seven months. The milestone *Full System Ready* is placed at Month 6, marking the point at which all subsystems are integrated and operational, leaving Month 7 dedicated to validation experiments and report writing.

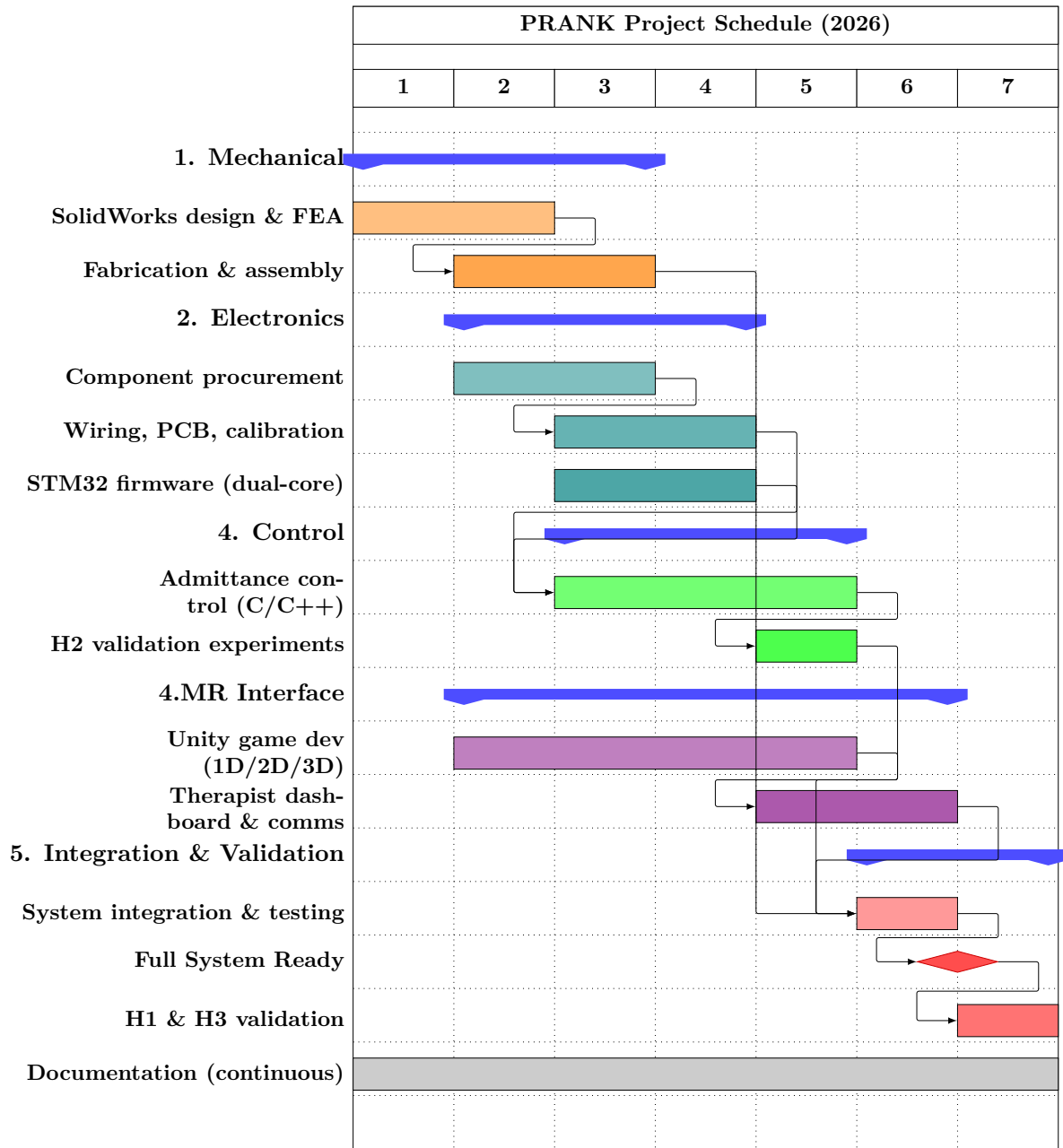


Figure 8.1: Seven-month project schedule for PRANK. Colours indicate stage: orange (mechanical), teal (electronics), green (control), purple (MR game), red (integration/validation), grey (documentation).

Chapter 9

PROJECT RESOURCES

This chapter outlines the technical requirements, material resources, and estimated costs associated with the development of the PRANK system (Parallel Robot for ANKle rehabilitation). It includes a detailed breakdown of mechanical components, electronic subsystems, software tools, and validation instruments necessary to construct and implement the rehabilitation platform. Additionally, the chapter provides an initial estimation of development time and budget, supporting the project's feasibility and helping guide logistical and financial planning. These projections will serve as a reference for resource allocation during the execution phase of the project.

Table 9.1: Preliminary Resource and Cost Estimation for PRANK

Category	Description	Estimated Cost (USD)
3D Printing	PLA filament, high-resolution printing (frame and joint parts)	180
Electronic Components	Microcontroller (STM32), force sensor, IMUs, wiring, connectors	500
Motors	3 $\times$ Brushless DC motors with encoders	240
Motor Drivers	Compatible drivers with current control	90
Mechanical parts	Bearings, couplings, screws, aluminum parts	100
VR Headset	Oculus/Meta Quest 2 (or equivalent)	300
Software Licenses	Unity Pro/Unreal (if needed), MATLAB/Simulink (edu license)	0–100
Validation Tools	Access to optoelectronic motion capture lab	50
Personal Labor	Estimation of 120 hours $\times$ \$10/hr (development + testing)	1,200
Total Budget		2,760

Bibliography

- [1] W. S. Organization, “Global stroke fact sheet 2025,” *PubMed Central*, 2025, Accessed March 29, 2026.
- [2] Recavar, *Estadísticas acv: Datos de colombia y el mundo*, <https://www.recavar.org/estadisticas-acv>, Accedido el 29 de marzo de 2026, 2026.
- [3] M. Clinic, “Foot drop: Causes,” 2025, Accessed March 29, 2026.
- [4] C. Clinic, “Foot drop: Symptoms and causes,” 2025, Accessed March 29, 2026.
- [5] N. I. of Neurological Disorders and Stroke, “Foot drop information page,” 2025, Accessed March 29, 2026.
- [6] MedlinePlus, “Foot drop,” 2025, Accessed March 29, 2026.
- [7] J. H. Medicine, “Foot drop,” 2025, Accessed March 29, 2026.
- [8] C. Clinic, “Foot drop: Symptoms and causes,” 2025, Accessed March 29, 2026.
- [9] M. Clinic, “Foot drop: Causes,” 2025, Accessed March 29, 2026.
- [10] I. D. Federation, “Idf diabetes atlas, 10th edition,” 2025, Accessed March 29, 2026.
- [11] J. D. Stewart, “Foot drop: Where, why and what to do,” *Practical Neurology*, vol. 8, no. 3, pp. 158–169, 2008. DOI: 10.1136/jnnp.2008.149393.
- [12] A. I. R. Kottink, L. J. L. Oostendorp, J. H. Buurke, A. V. Nene, H. J. Hermens, and M. J. IJzerman, “The orthotic effect of functional electrical stimulation on the improvement of walking in stroke patients with a dropped foot: A systematic review,” *Artificial Organs*, vol. 28, no. 6, pp. 577–586, 2004. DOI: 10.1111/j.1525-1594.2004.07310.x.
- [13] S. Truijen et al., “Effect of home-based virtual reality training and telerehabilitation on balance in individuals with parkinson disease, multiple sclerosis, and stroke: A systematic review and meta-analysis,” *Neurological Sciences*, vol. 43, pp. 2995–3006, 5 2022, ISSN: 1590-3478. DOI: 10.1007/s10072-021-05855-2.
- [14] Y. H. Tsoi and S. Q. Xie, “Design and control of a parallel robot for ankle rehabilitation,” *International Journal of Intelligent Systems Technologies and Applications*, vol. 8, pp. 100–113, 1-4 Dec. 2009, doi: 10.1504/IJISTA.2010.030193, ISSN: 1740-8865. DOI: 10.1504/IJISTA.2010.030193.
- [15] J. M. Morris, “Biomechanics of the foot and ankle,” *Clinical Orthopaedics and Related Research®*, pp. 10–17, 122 1977.
- [16] P.-Y. Lin, Y.-R. Yang, S.-J. Cheng, and R.-Y. Wang, “The relation between ankle impairments and gait velocity and symmetry in people with stroke,” *Archives of physical medicine and rehabilitation*, vol. 87, pp. 562–568, 4 2006.

- [17] R. Baker, *Measuring walking: Handbook of clinical gait analysis*, 2013. [Online]. Available: <https://search.ebscohost.com/login.aspx?direct=true&db=e000xww&AN=579716&site=eds-live>.
- [18] V. Medved, "History of the study of human locomotion and elements of current research methodology bt - measurement and analysis of human locomotion," in V. Medved, Ed. Springer International Publishing, 2021, pp. 13–37, ISBN: 978-3-030-79685-3. DOI: 10.1007/978-3-030-79685-3_2.
- [19] F. Stevens, N. J. Weerkamp, and J. W. L. Cals, "Foot drop," *BMJ: British Medical Journal*, vol. 350, 2015, ISSN: 09598138, 17561833.
- [20] F. Alnajjar, R. Zaier, S. Khalid, and M. Gochoo, "Trends and technologies in rehabilitation of foot drop: A systematic review," *Expert Review of Medical Devices*, vol. 18, pp. 31–46, 1 2021, Export Date: 03 March 2022; Cited By: 5. DOI: 10.1080/17434440.2021.1857729.
- [21] P. K. Jamwal, S. Xie, and K. C. Aw, "Kinematic design optimization of a parallel ankle rehabilitation robot using modified genetic algorithm," *Robotics and Autonomous Systems*, vol. 57, pp. 1018–1027, 10 2009, ISSN: 0921-8890. DOI: <https://doi.org/10.1016/j.robot.2009.07.017>.
- [22] P. K. Jamwal, S. Hussain, and S. Q. Xie, "Three-stage design analysis and multicriteria optimization of a parallel ankle rehabilitation robot using genetic algorithm," *IEEE Transactions on Automation Science and Engineering*, vol. 12, pp. 1433–1446, 4 2015, ISSN: 1558-3783. DOI: 10.1109/TASE.2014.2331241.
- [23] P. K. Jamwal, S. Hussain, M. H. Ghayesh, and S. V. Rogozina, "Impedance control of an intrinsically compliant parallel ankle rehabilitation robot," *IEEE Transactions on Industrial Electronics*, vol. 63, pp. 3638–3647, 6 2016, ISSN: 1557-9948. DOI: 10.1109/TIE.2016.2521600.
- [24] C. Wang, Y. Fang, and S. Guo, "Multi-objective optimization of a parallel ankle rehabilitation robot using modified differential evolution algorithm," *Chinese Journal of Mechanical Engineering*, vol. 28, pp. 702–715, 4 2015, ISSN: 2192-8258. DOI: 10.3901/CJME.2015.0416.062.
- [25] M. Zhang et al., "Adaptive patient-cooperative control of a compliant ankle rehabilitation robot (carr) with enhanced training safety," *IEEE Transactions on Industrial Electronics*, vol. 65, pp. 1398–1407, 2 2018, ISSN: 1557-9948. DOI: 10.1109/TIE.2017.2733425.
- [26] M. Zhang, A. McDaid, A. J. Veale, Y. Peng, and S. Q. Xie, "Adaptive trajectory tracking control of a parallel ankle rehabilitation robot with joint-space force distribution," *IEEE Access*, vol. 7, pp. 85 812–85 820, 2019, ISSN: 2169-3536. DOI: 10.1109/ACCESS.2019.2925182.
- [27] Q. Miao, M. Zhang, C. Wang, and H. Li, "Towards optimal platform-based robot design for ankle rehabilitation: The state of the art and future prospects," *Journal of Healthcare Engineering*, vol. 2018, p. 1 534 247, 1 Jan. 2018, ISSN: 2040-2295. DOI: <https://doi.org/10.1155/2018/1534247>.

- [28] J. Li et al., “Mechanical design and performance analysis of a novel parallel robot for ankle rehabilitation,” *Journal of Mechanisms and Robotics*, vol. 12, 5 Apr. 2020, ISSN: 1942-4302. DOI: 10.1115/1.4046511.
- [29] P. S. Nazar and P. P. Pott, “Ankle rehabilitation robotic systems for domestic use – a systematic review,” vol. 8, pp. 65–68, 2 2022. DOI: doi:10.1515/cdbme-2022-1018.
- [30] J. Li, W. Fan, M. Dong, and X. Rong, “Implementation of passive compliance training on a parallel ankle rehabilitation robot to enhance safety,” *Industrial Robot: the international journal of robotics research and application*, vol. 47, pp. 747–755, 5 Jan. 2020, ISSN: 0143-991X. DOI: 10.1108/IR-02-2020-0040.
- [31] X. Zhai et al., “Effects of robot-aided rehabilitation on the ankle joint properties and balance function in stroke survivors: A randomized controlled trial,” *Frontiers in Neurology*, vol. 12, p. 719305, 2021. DOI: 10.3389/fneur.2021.719305.
- [32] K. E. Zelik and E. C. Honert, “Ankle and foot power in gait analysis: Implications for science, technology and clinical assessment,” *Journal of Biomechanics*, vol. 75, pp. 1–12, 2018.
- [33] B. Zhong, K. Guo, H. Yu, and M. Zhang, “Toward gait symmetry enhancement via a cable-driven exoskeleton powered by series elastic actuators,” *IEEE Robotics and Automation Letters*, vol. 7, pp. 786–793, 2 2022, Export Date: 03 March 2022; Cited By: 0. DOI: 10.1109/LRA.2021.3130639.
- [34] D. Gomez-Vargas, M. J. Pinto-Betnal, F. Ballen-Moreno, M. Munera, and C. A. Cifuentes, *Therapy with t-flex ankle-exoskeleton for motor recovery: A case study with a stroke survivor*, 2020. DOI: 10.1109/BioRob49111.2020.9224277. [Online]. Available: <https://search.ebscohost.com/login.aspx?direct=true&db=edsee&AN=edsee.9224277&site=eds-live>.
- [35] C. L. Brockett and G. J. Chapman, “Biomechanics of the ankle,” *Orthopaedics and Trauma*, vol. 30, no. 3, pp. 232–238, 2016. DOI: 10.1016/j.mporth.2016.04.015.
- [36] D. Schneck and J. Bronzino, *Biomechanics: Principles and Applications*. CRC Press, 2002, ISBN: 9781420040029.
- [37] D. A. Winter, *Biomechanics and Motor Control of Human Movement*, 4th. Hoboken, NJ: John Wiley & Sons, 2009, ISBN: 978-0-470-39818-0.
- [38] K. Shamaei, G. S. Sawicki, and A. M. Dollar, “Estimation of quasi-stiffness and propulsive work of the human ankle in the stance phase of walking,” *PLOS ONE*, vol. 8, no. 3, e59935, 2013. DOI: 10.1371/journal.pone.0059935.
- [39] A. H. Hansen, D. S. Childress, S. C. Miff, S. A. Gard, and K. P. Mesplay, “The human ankle during walking: Implications for design of biomimetic ankle prostheses,” *Journal of Biomechanics*, vol. 37, no. 10, pp. 1467–1474, 2004. DOI: 10.1016/j.jbiomech.2004.01.017.
- [40] J. Hicks and O. Team, *Simulation-based design to prevent ankle injuries*, <https://opensimconfluence.atlassian.net/wiki/spaces/OpenSim/pages/53088618/Simulation-Based+Design+to+Prevent+Ankle+Injuries>, Accessed November 2025, 2024.

- [41] H. Malakoutikhah, C. de Cesar Netto, E. Madenci, and L. D. Latt, “Evaluation of assumptions in foot and ankle biomechanical models,” *Clinical Biomechanics*, vol. 100, p. 105807, 2022. DOI: 10.1016/j.clinbiomech.2022.105807.
- [42] Physiopedia, *Role of neuroplasticity in neuro-rehabilitation*, https://www.physio-pedia.com/Role_of_Neuroplasticity_in_Neuro-rehabilitation, Accessed November 2025, 2024.
- [43] G. Morone et al., “Robot- and technology-boosting neuroplasticity-dependent motor-cognitive functional recovery: Looking towards the future of neurorehabilitation,” *Brain Sciences*, vol. 13, no. 12, p. 1687, 2023. DOI: 10.3390/brainsci13121687.
- [44] W. Heng, S. Solomon, and W. Gao, “Flexible electronics and devices as human-machine interfaces for medical robotics,” *Advanced Materials*, 2021, Accessed November 2025.
- [45] C. A. Cifuentes, J. F. Veneman, E. Rocon, and C. Rodriguez-Guerrero, “Interfacing humans and machines for rehabilitation and assistive devices,” *Frontiers in Robotics and AI*, vol. 8, 2021. DOI: 10.3389/frobt.2021.796431.
- [46] S. R. AbilityLab, *Rehabilitation measures database*, <https://www.sralab.org/rehabilitation-measures>, Accessed November 2025, 2024.

ANNEXES